REVERSE ENGINEERING – ACTIVITY REPORT

Product: Logitech M331 Silent Plus Wireless Mouse

PART I – STUDYING THE PRODUCT

1. External Features

Physical Construct

- **Shape:** Asymmetric, contoured right-hand body, approx. 106 × 68 × 39 mm.
- **Top:** Single moulded button cover with left and right buttons, divided by the scroll-wheel slot; rubber scroll wheel in the centre.
- **Sides:** Rubber side grips for a secure, comfortable hold.
- **Bottom:** ON/OFF slide switch, optical sensor window, low-friction feet, and a removable battery door covering the AA battery tube and the USB receiver storage.
- **Accessory:** USB nano receiver (2.4 GHz) that plugs into the computer.
- **Assembly:** Plastic parts held by snap hooks plus only two screws (one hidden under the label in the battery well).

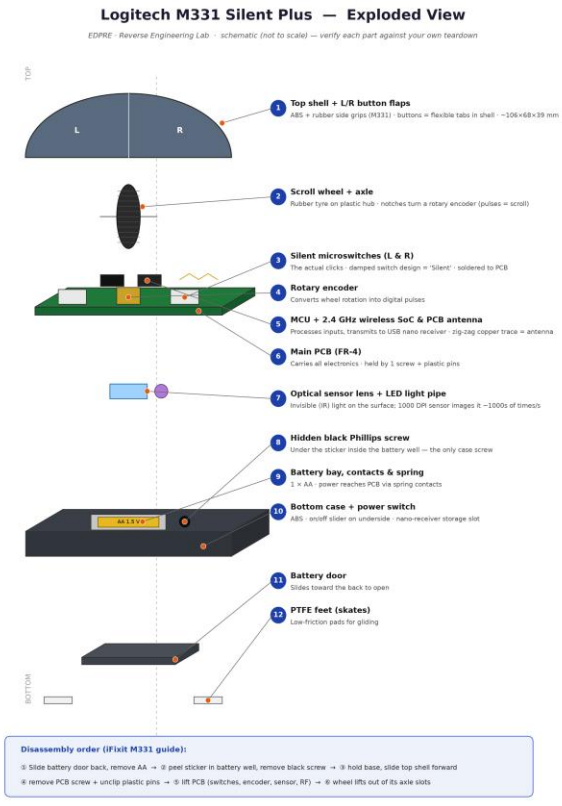


Fig. A – Exploded view of the M331 (schematic).

Subsystems Used

- Power subsystem
- Optical tracking subsystem
- Input subsystem (buttons and scroll wheel)
- Processing & wireless communication subsystem
- Mechanical / enclosure subsystem

Description of Subsystems

- **Power:** one 1.5 V AA battery, spring contacts, ON/OFF switch and a regulator circuit supply all electronics.
- **Optical tracking:** an infrared LED, lens and image sensor detect how far and in which direction the mouse moves.
- **Input:** silent left/right switches, a wheel-click switch and a scroll-wheel encoder capture user commands.
- **Processing & wireless:** a microcontroller with a 2.4 GHz radio collects all inputs and transmits them to the USB receiver.
- **Mechanical:** the housing, grips, feet and battery door protect the parts and give comfort.

(Detailed explanation of each subsystem is given in Part III, Section 4.)

Modes of Operation

- **OFF mode:** slide switch OFF — no power is drawn (used for travel/storage).
- **Active mode:** sensor tracks movement continuously; clicks and scrolls are sent to the computer.
- **Idle / sleep mode:** after a period of no use, the electronics enter low-power sleep to save battery.
- **Wake-up:** movement or a click wakes the mouse instantly and it reconnects to the receiver.

2. Purpose of the Product

The M331 is a wireless pointing device used to control the on-screen cursor, select and open items, and scroll through content — with the specific purpose of doing this quietly, so it can be used in shared or quiet spaces without click noise.

3. User Needs Addressed

User need	How the M331 addresses it
Quiet working (libraries, offices, at night)	Silent switches reduce click noise by over 90%
Freedom from cables	2.4 GHz wireless with USB receiver, up to 10 m range
Low maintenance	Up to 18 months on one AA battery; no cleaning (optical)
Comfort over long use	Contoured right-hand shape and rubber grips
Portability	Compact size; receiver can be stored inside; ON/OFF switch
Ease of use	Plug-and-play on Windows, macOS, Linux — no driver needed
Affordability and reliability	Budget price; no moving tracking parts to wear out

PART II – PRODUCT ORIGIN TO LATEST INVENTIONS

1. Difference in Technology Across Versions

The computer mouse has evolved from a wooden mechanical prototype to today's high-precision wireless optical devices. Each stage solved the main drawback of the previous one.

1.1 Early Stage (1964–1968): Mechanical wheel mouse

- **Origin:** Invented by Douglas Engelbart with engineer Bill English at the Stanford Research Institute (1964); publicly shown in the 1968 'Mother of All Demos'; patented in 1970 as an 'X-Y position indicator'.
- **Technology:** A wooden box with two perpendicular metal wheels; each wheel turned a potentiometer to measure X and Y motion. One button.
- **Improvement:** First practical pointing device — direct on-screen selection instead of typed commands.
- **Drawbacks:** Bulky, wired, could only roll smoothly along one axis at a time, needed a lot of desk space.

1.2 First Generation (1972 – late 1990s): Ball (opto-mechanical) mouse

- **Technology:** Bill English (Xerox PARC, 1972) replaced the wheels with a rubber ball. The ball turns two rollers (X and Y); each roller spins a slotted disc between an LED and a photo-detector, producing pulses. Used on the Xerox Alto/Star and popularised by the Apple Lisa (1983) and Macintosh (1984).
- **Improvement:** Smooth motion in any direction, cheap to manufacture, became the standard PC input device.
- **Drawbacks:** Dust and dirt collected on the ball and rollers (needed regular cleaning), required a mouse pad, low resolution, mechanical wear.

1.3 Second Generation (1995 – early 2000s): Scroll wheel and LED optical mouse

- **Technology:** The scroll wheel was added in the mid-1990s (e.g., Microsoft IntelliMouse, 1996). From 1999, LED optical mice (e.g., Microsoft IntelliMouse Explorer) used a tiny camera sensor that photographs the surface thousands of times per second and compares images to detect movement.
- **Improvement:** No moving tracking parts → no cleaning, much higher precision and reliability, works on most surfaces without a pad.
- **Drawbacks:** Poor tracking on glass or glossy surfaces; still wired.

1.4 Third Generation (2000s): Wireless and laser mice

- **Technology:** Early cordless mice used infrared/RF; 2.4 GHz radio with small USB receivers became standard. Laser sensors (e.g., Logitech MX1000, 2004) gave higher resolution and worked on more surfaces. Logitech's Unifying receiver (2009) let one receiver serve several devices.
- **Improvement:** Freedom from cables, longer range, better tracking.
- **Drawbacks:** Batteries needed; early wireless had lag and interference.

1.5 Latest Generation (2010s – present): Smart, efficient and ergonomic mice

- **Technology:** Very-high-DPI sensors (tens of thousands of DPI in gaming mice), tracking on glass (Logitech Darkfield), low-latency wireless, Bluetooth multi-device use, rechargeable USB-C batteries, ultra-low-power chips giving 1–2 years on one AA cell, silent switches (the M330/M331 Silent Plus, 2016), ergonomic and vertical designs.
- **Improvement:** Long battery life, quiet operation, comfort, multi-device productivity.

- **Drawbacks:** Higher cost for premium models; more electronic waste from batteries and non-repairable designs.

2. Current Status – Core Smart Technologies Used in the Product (M331)

Technology	What it does in the M331
Optical tracking (invisible IR light)	1000 DPI optical sensor; no visible glow; reliable on most desk surfaces
2.4 GHz wireless + USB nano receiver	Plug-and-play connection, range up to 10 m (Logitech spec)
SilentTouch switch technology	Reduces click noise by over 90% (Logitech spec) while keeping the same click feel
Ultra-low-power electronics	Up to 18 months on one AA battery (Logitech spec); auto-sleep when idle + ON/OFF switch
Ergonomic design	Contoured right-hand shape with rubber side grips; rubber scroll wheel
Cross-platform HID	Works on Windows, macOS, Linux without special drivers

3. Engineering Importance of This Evolution

- **Mechanical → optical → electronic:** moving parts were replaced by sensors and chips, giving higher reliability and precision.
- **Integration (VLSI):** the optical sensor is a CMOS image sensor with an on-chip DSP, and the MCU + radio are combined in one SoC — many functions on one chip lowers cost, size and power.
- **Low-power design:** efficient chips and sleep modes stretch one AA cell to over a year.
- **Wireless communication:** 2.4 GHz digital radio links replaced cables.
- **Human factors:** ergonomics and noise reduction show engineering for the user, not just function.
- **Design for manufacturing:** snap-fit housings and very few screws (only two in the M331) reduce assembly cost.

PART III – INTERNAL COMPONENTS & WORKING FLOW

1. Introduction of the Product

The Logitech M331 Silent Plus (released 2016; sold as the M330 in some regions) is a right-handed wireless optical mouse with silent clicks. It has three inputs (left click, right click, scroll wheel with wheel-click), runs on one AA battery and connects to a computer through a 2.4 GHz USB nano receiver. Approximate size: 106 × 68 × 39 mm.

2. Photographs of the Internal Components



Fig. 1 – Fully disassembled M331: top button cover, upper shell, base with battery tube, battery door, scroll wheel, PCB, lens, screws, spring, label.

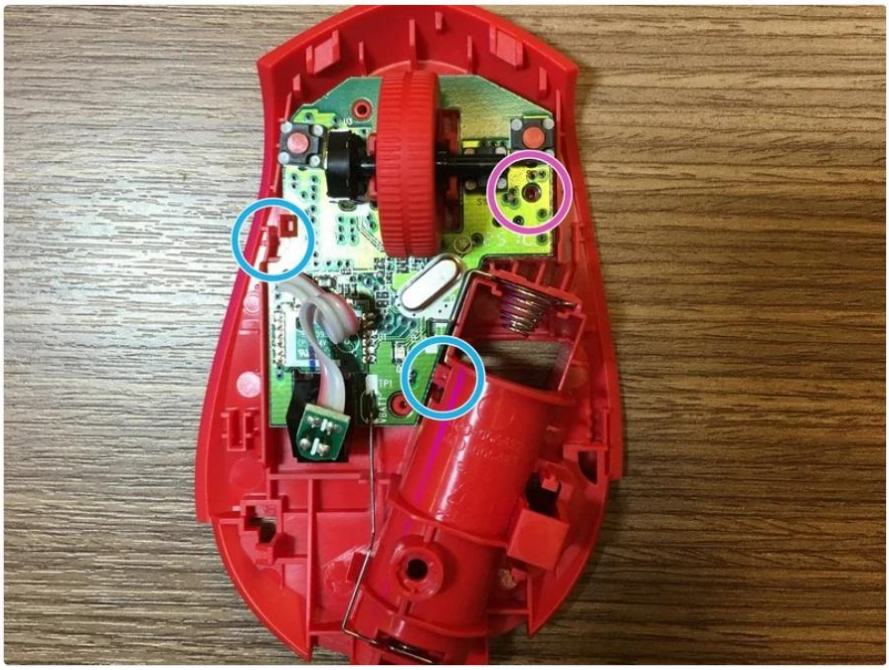


Fig. 2 – Main PCB inside the base: L/R silent switches, scroll wheel and encoder, crystal oscillator, wiring, battery tube and spring.

Photo source: iFixit, 'Logitech M331 Disassembly' guide (CC BY-NC-SA).

3. Internal Block Diagram (Functional View)

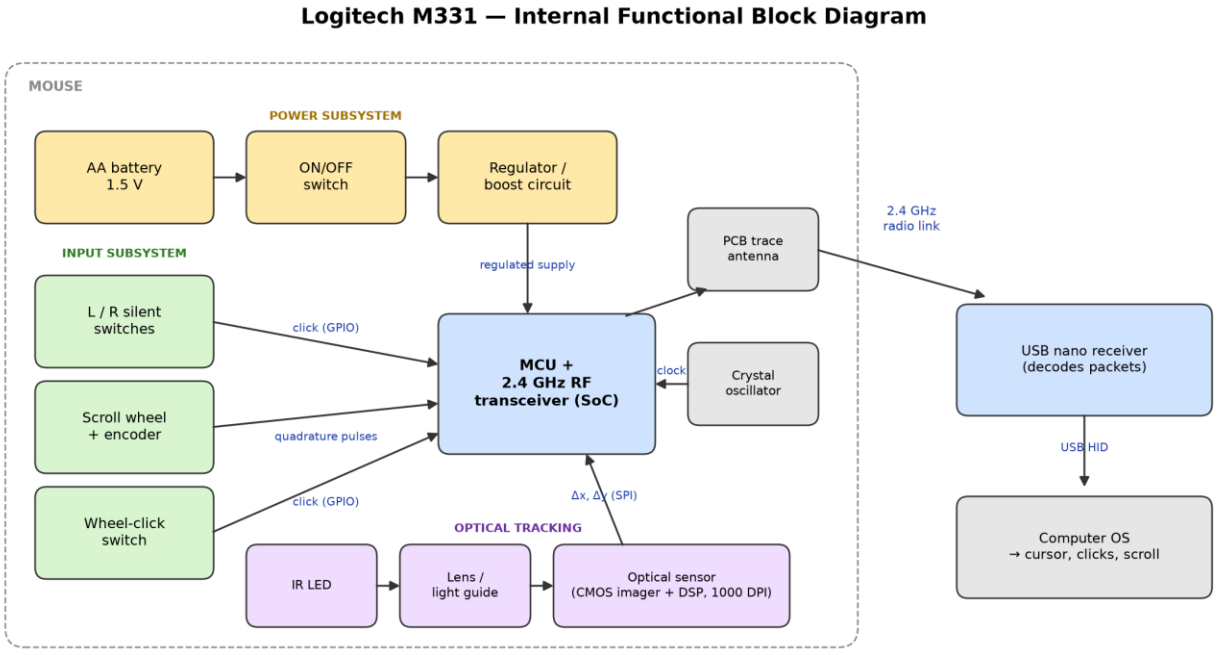


Fig. 3 – Functional block diagram of the M331.

4. Internal Subsystems with Detailed Explanation

4.A Power subsystem (AA battery, ON/OFF switch, contacts/spring, regulator/boost circuit)

4.1 Function (why it is used): Supplies stable electrical power to all electronics.

4.2 Working principle: The 1.5 V AA cell connects through spring contacts and the slide switch; a regulator/boost circuit converts it to the steady voltage the chips need.

4.3 Characteristics: Single AA cell; up to 18 months battery life (Logitech spec); auto-sleep reduces consumption.

4.4 Role in this product: Without it nothing works; the efficient design is why the mouse runs for over a year on one battery.

4.B Optical tracking subsystem (IR LED, lens/light guide, optical sensor)

4.1 Function (why it is used): Detects the movement of the mouse across the desk.

4.2 Working principle: The LED lights the desk surface with invisible infrared light; the lens focuses the surface texture onto a tiny CMOS image sensor that takes thousands of pictures per second. Its built-in DSP compares consecutive images to compute the shift Δx, Δy.

4.3 Characteristics: Optical, 1000 DPI; no moving parts; works on most non-glass surfaces.

4.4 Role in this product: Converts physical hand motion into cursor motion — the core function of a mouse.

4.C Input subsystem (L/R silent switches, wheel-click switch, scroll wheel + encoder)

4.1 Function (why it is used): Captures clicks and scrolling.

4.2 Working principle: Pressing a button closes a switch (a digital 0/1 signal to the MCU). Turning the wheel rotates an encoder that produces two offset pulse trains (quadrature); the order of the pulses tells the direction of scroll.

4.3 Characteristics: Silent switches cut click noise by over 90% (Logitech spec); rubber wheel with notched feel.

4.4 Role in this product: Lets the user select, open menus and scroll.

4.D Processing & wireless subsystem (MCU + 2.4 GHz RF SoC, crystal oscillator, PCB antenna, USB receiver)

4.1 Function (why it is used): Combines all inputs and sends them to the computer.

4.2 Working principle: The microcontroller reads the sensor (over SPI) and the switches/encoder, debounces clicks, packs the data into a report and transmits it by 2.4 GHz radio through the zig-zag PCB antenna. The crystal provides the precise clock frequency the radio needs. The USB receiver decodes the packets and presents them to the computer as a standard HID mouse.

4.3 Characteristics: 2.4 GHz digital radio; range up to 10 m (Logitech spec); plug-and-play (no driver).

4.4 Role in this product: The 'brain' and the wireless link of the product.

4.E Mechanical / enclosure subsystem (top button cover, upper shell with rubber grips, base with battery tube, battery door, snap hooks, 2 screws, feet)

4.1 Function (why it is used): Holds and protects all parts and gives the ergonomic shape.

4.2 Working principle: Moulded plastic parts clip together with snap hooks; only one case screw (under the label in the battery well) and one PCB screw are used. Button flaps press the switches below them; low-friction feet let the mouse glide.

4.3 Characteristics: ABS plastic housing, rubber side grips, ~106 × 68 × 39 mm.

4.4 Role in this product: Determines comfort, durability and ease of assembly/repair.

5. Complete Working Flow (Step-by-Step)

Step 1: User slides the ON/OFF switch to ON → battery powers the regulator → MCU, radio and sensor start; the mouse links to its USB receiver.

Step 2: The IR LED lights the desk surface; the lens focuses the surface image onto the optical sensor.

Step 3: The sensor captures images thousands of times per second; its DSP compares consecutive images and calculates movement Δx, Δy.

Step 4: The MCU reads Δx, Δy from the sensor (SPI) and checks the switches and the wheel encoder.

Step 5: When a button is pressed, its switch closes; the MCU debounces it and records a click.

Step 6: When the wheel turns, the encoder's quadrature pulses tell the MCU the scroll amount and direction.

Step 7: The MCU packs movement, buttons and scroll into a data packet and sends it by 2.4 GHz radio through the PCB antenna (timed by the crystal oscillator).

Step 8: The USB nano receiver decodes the packet and sends it to the computer as a standard USB HID report.

Step 9: The operating system moves the cursor, performs clicks or scrolls the page.

Step 10: If the mouse is not used, it enters a low-power sleep mode to save battery, and wakes on movement or click.

6. Components List – Bill of Materials

No.	Component	Qty	Material / Type
1	Top button cover (L/R buttons, wheel slot)	1	ABS plastic
2	Upper shell with side grips	1	ABS + rubber
3	Base / bottom case with battery tube	1	ABS plastic
4	Battery door	1	ABS plastic
5	Scroll wheel (rubber tyre)	1	Plastic hub + rubber
6	Wheel encoder	1	Rotary encoder
7	Main PCB	1	FR-4 fibreglass, copper traces
8	Silent tactile switches (L, R)	2	Damped tactile switch
9	Wheel-click switch	1	Tactile switch
10	MCU + 2.4 GHz RF transceiver	1	SoC IC

11	Crystal oscillator	1	Quartz crystal (metal can)
12	PCB trace antenna	1	Copper trace on PCB
13	Optical sensor	1	CMOS image sensor + DSP IC
14	IR LED	1	Infrared LED
15	Lens / light guide	1	Clear plastic (polycarbonate/acrylic)
16	ON/OFF slide switch	1	Slide switch
17	Battery contacts + spring	1	Spring steel, plated
18	Small sub-board with wires	1	PCB + wires
19	Screws	2	Steel, Phillips head
20	Feet (skates)	set	Low-friction pads
21	Label sticker	1	Paper/plastic label
22	USB nano receiver	1	USB dongle with RF chip
23	AA battery	1	1.5 V alkaline

7. Component Ratings & Specifications

Item	Rating / Specification	Source
Battery	1 × AA, 1.5 V; up to 18 months life	Logitech
Wireless	2.4 GHz via USB receiver; range up to 10 m	Logitech
Click noise	Over 90% reduction vs standard clicks	Logitech
Sensor	Optical, invisible light, 1000 DPI	Product review (Techspin)
Size	~106 × 68 × 39 mm	Product review (Techspin)
Compatibility	Windows, macOS, Linux (USB HID)	Logitech
Crystal frequency	Read the marking on the metal can	To be measured
Switch life / IC part numbers	Not published — read chip markings	To be measured

8. Component Function Table

Component Name	Function	Specifications
Optical sensor	Images the surface, computes Δx/Δy	1000 DPI, CMOS + DSP
IR LED + lens	Illuminates and focuses the surface	Infrared (invisible)
MCU + RF SoC	Reads inputs, builds and transmits reports	2.4 GHz, low power
Crystal oscillator	Precise clock for MCU/radio	Quartz (value on can)
PCB antenna	Radiates the 2.4 GHz signal	Copper trace
L/R silent switches	Register left/right clicks quietly	>90% quieter clicks
Scroll wheel + encoder	Scrolling with direction sensing	Rubber wheel, quadrature
Wheel-click switch	Middle click	Tactile switch
AA battery	Power source	1.5 V, up to 18 months
ON/OFF switch	Cuts power when not in use	Slide switch
USB nano receiver	Receives radio data, connects via USB	USB HID, 10 m range
Housing (shell, base, door)	Protection, ergonomics, assembly	ABS + rubber grips

Feet	Smooth gliding	Low-friction pads

9. Simulation of the Product's Working (Tinkercad Circuits)

The internal working of the M331 was simulated in Autodesk Tinkercad Circuits. Tinkercad has no optical sensor or 2.4 GHz radio, so the mouse's inputs are imitated with simple components and an Arduino Uno plays the role of the mouse's microcontroller. The Serial Monitor shows the movement/button report that the real mouse would transmit to the computer.

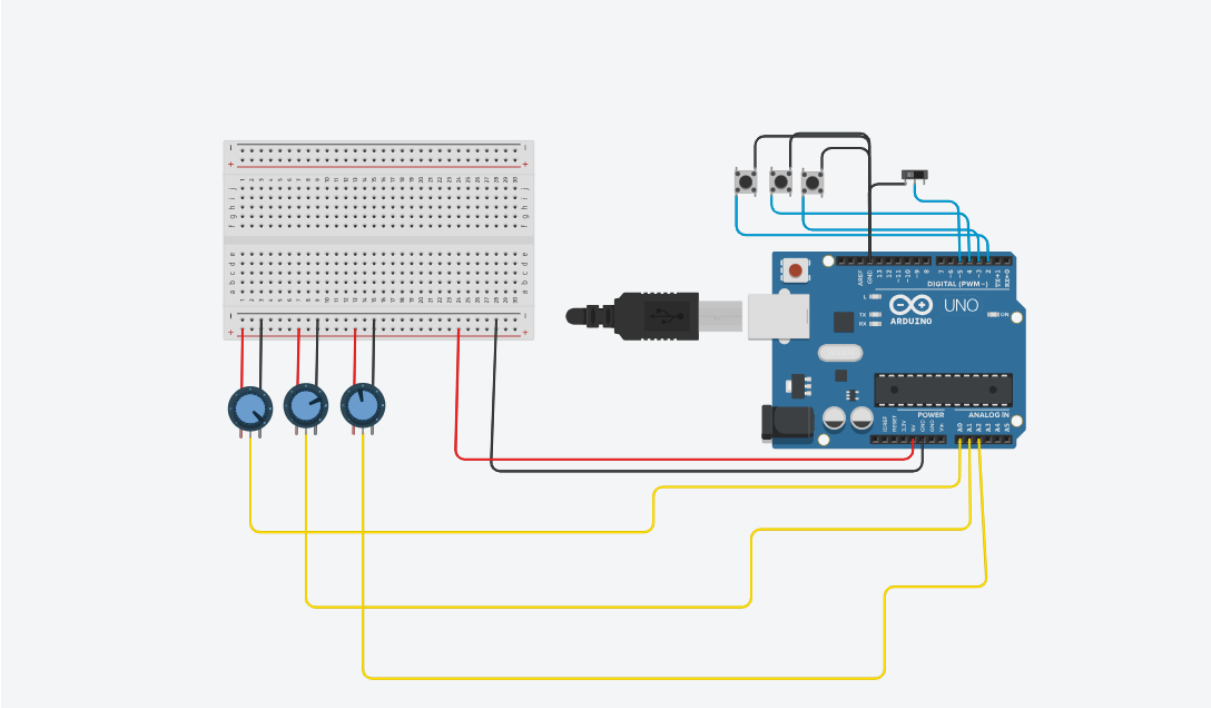


Fig. 4 – Tinkercad circuit: Arduino Uno (MCU), 3 potentiometers, 3 pushbuttons, slide switch, breadboard power rails.

M331 part	Simulated by	Arduino pin	Output in report
Optical sensor – X movement	Potentiometer 1	A0	dx
Optical sensor – Y movement	Potentiometer 2	A1	dy
Scroll wheel + encoder	Potentiometer 3	A2	scroll
Left silent switch	Pushbutton 1	D2	L
Right silent switch	Pushbutton 2	D3	R
Wheel-click switch	Pushbutton 3	D4	M
ON/OFF power switch	Slide switch	D5	LED on / reports stop
MCU + radio link	Arduino Uno + Serial Monitor	—	REPORT lines

9.1 How the simulation works

- **Power:** Arduino 5V and GND feed the breadboard rails, which power the potentiometers (stands in for the AA battery and regulator).

- **Movement:** each potentiometer is a voltage divider; analogRead() converts its wiper voltage to 0–1023. The code reports the change since the last reading (dx = (x – lastX)/8), just as the optical sensor reports change in position. Dividing by 8 ignores small noise.
- **Buttons and switch:** pins use INPUT_PULLUP, so they read HIGH normally and LOW when the button/switch connects them to GND — no external resistors needed.
- **Main loop:** about 20 times per second the code checks the power switch (OFF → LED off, nothing sent = sleep), reads all inputs and, if anything changed, prints one report line — the same read → pack → send cycle the M331's chip performs.

9.2 Results

Sliding the switch ON lit the Arduino's L LED and enabled reports; turning the knobs produced dx, dy and scroll values; pressing each button produced L=1, R=1 or M=1; switching OFF stopped all reports. The simulation therefore reproduces the mouse's input → processing → report flow.

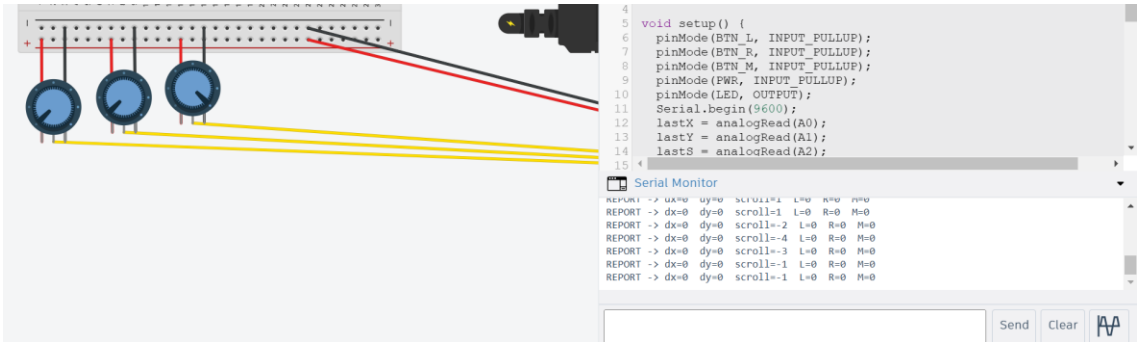


Fig. 5 – Serial Monitor: scroll values while turning potentiometer 3.

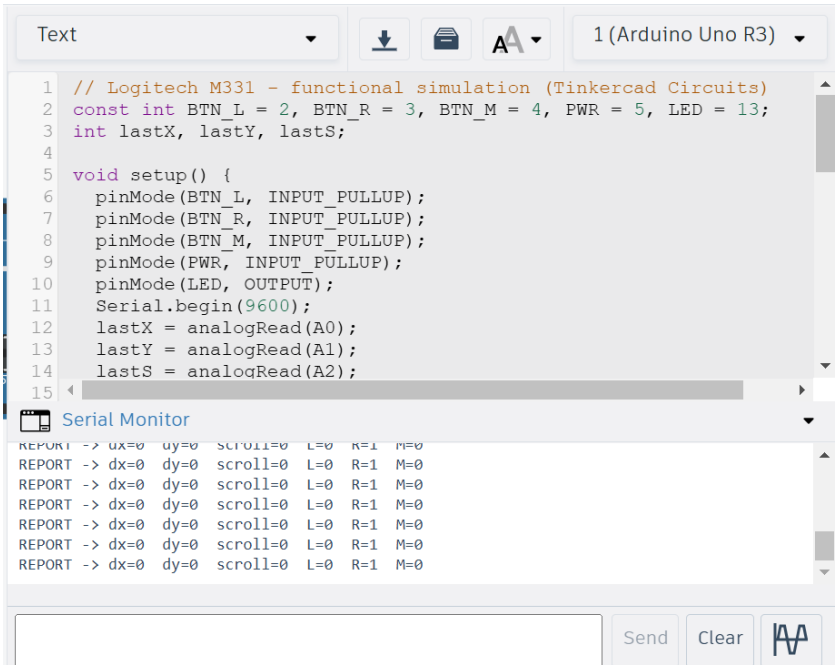


Fig. 6 – Code (Text view) and Serial Monitor showing R=1 while the right button is held.

9.3 Program code

```
// Logitech M331 - functional simulation (Tinkercad Circuits)
const int BTN_L = 2, BTN_R = 3, BTN_M = 4, PWR = 5, LED = 13;
int lastX, lastY, lastS;

void setup() {
  pinMode(BTN_L, INPUT_PULLUP);
  pinMode(BTN_R, INPUT_PULLUP);
  pinMode(BTN_M, INPUT_PULLUP);
  pinMode(PWR, INPUT_PULLUP);
  pinMode(LED, OUTPUT);
  Serial.begin(9600);
  lastX = analogRead(A0);
  lastY = analogRead(A1);
  lastS = analogRead(A2);
  Serial.println("M331 simulation ready - slide switch ON to start");
}

void loop() {
  if (digitalRead(PWR) == HIGH) {     // switch OFF
    digitalWrite(LED, LOW);
    delay(200);
    return;
  }
  digitalWrite(LED, HIGH);            // switch ON

  int x = analogRead(A0), y = analogRead(A1), s = analogRead(A2);
  int dx = (x - lastX) / 8;
  int dy = (y - lastY) / 8;
  int scroll = (s - lastS) / 64;
  if (dx != 0) lastX = x;
  if (dy != 0) lastY = y;
  if (scroll != 0) lastS = s;

  bool L = digitalRead(BTN_L) == LOW;
  bool R = digitalRead(BTN_R) == LOW;
  bool M = digitalRead(BTN_M) == LOW;

  if (dx || dy || scroll || L || R || M) {
    Serial.print("REPORT -> dx=");  Serial.print(dx);
    Serial.print("  dy=");          Serial.print(dy);
    Serial.print("  scroll=");      Serial.print(scroll);
    Serial.print("  L=");           Serial.print(L);
    Serial.print("  R=");           Serial.print(R);
    Serial.print("  M=");           Serial.println(M);
  }
  delay(50);
}
```

Sources: Tinkercad Circuits (Autodesk); Logitech M330/M331 product page; iFixit 'Logitech M331 Disassembly'; Techspin M330/M331 review; general history of the computer mouse.